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Jennifer Fill, Carolina Baruzzi & Raelene Crandall

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Perennial grasses with different life histories maintain population stability under fire in North and Central American pine savannas

Jennifer M. Fill^{1*}, Carolina Baruzzi^{1,2}, and Raelene M. Crandall¹

¹School of Forest, Fisheries, and Geomatics Sciences, University of Florida, 1745 McCarty Drive, Gainesville, FL, USA

²Current address: Department of Wildlife Ecology and Conservation, North Florida Research and Education Center, University of Florida, 155 Research Road, Quincy, FL, USA

*Correspondence, Jennifer M. Fill, jfill@ufl.edu

Abstract

Background: Although survival has been shown to be a strong driver of perennial grass population persistence in savanna understories, it is unclear how this pattern compares across species that display variation in reproductive strategies. We modeled the demography of four C4 perennial grass species with varying degrees of fire-stimulated flowering in the frequently burned pine savannas across North and Central America. We collected annual demographic data and modeled their population demography in annual transitions with and without fire using integral projection models.

Results: Although the species showed different reproductive patterns, all exhibited stable or growing populations across all transitions. Species balanced variable survival with relatively consistent reproduction and recruitment, whereas those with reproductive patterns more strongly affected by fire showed very high survival. Overall, survival was high across all species regardless of fire, and estimated recruitment for any species and year was low.

Conclusions: Survival and slow growth of adult individuals under frequent fire regimes are important for maintaining the persistence of perennial grass populations in these North and Central American savanna ecosystems. The perennial grasses we studied exhibit life history strategies typical of old-growth savanna

species with extensive and protected bud banks. The sensitivity of bud banks to fire varies among species, which may have influenced variable vital rates and patterns resulting from trade-offs between vegetative and sexual reproduction.

Keywords: Belize, demography, fire regime, grassland, survival, reproduction, wiregrass

Resumen

Antecedentes: Aunque la supervivencia ha demostrado ser un muy fuerte factor conducente en la persistencia de las poblaciones de pastos en el sotobosque de sabanas, no queda claro cómo este patrón se compara entre especies que presentan variaciones en estrategias reproductivas. Modelamos la demografía de cuatro especies con estrategias C4 con diferentes grados de estimulación de la floración debida al factor fuego, en sabanas de pino frecuentemente impactadas por incendios en América del Norte y América Central. Colectamos datos anuales sobre demografía y modulamos la demografía de las poblaciones en transiciones anuales con y sin fuegos, usando modelos de proyecciones integrales.

Resultados: Aunque las especies mostraron patrones reproductivos diferentes, todas exhibieron poblaciones estables o crecientes a través de todas las transiciones. Las especies balanceaban las variables de supervivencia con variables relativamente consistentes de reproducción y reclutamiento, mientras que aquellos con patrones reproductivos más afectados por el fuego mostraron una alta supervivencia. Y por sobre todo, la supervivencia fue alta en todas las especies independientemente del fuego, y el reclutamiento estimado para cualquier especie y años fue bajo.

Conclusiones: La supervivencia y el lento crecimiento de individuos adultos bajo regímenes de fuegos frecuentes son importantes para mantener la persistencia de las poblaciones de pastos perennes en esos ecosistemas de sabanas de América del Norte y América Central. Los pastos perennes que estudiamos exhibieron estrategias de formas de vida típicas de especies de sabana longevas, con bancos de meristemas de crecimiento extensos y protegidos. La sensibilidad de esos meristemas al fuego varió entre especies, lo que pudo haber influenciado los patrones y tasas vitales variables, siendo el resultado de una retroalimentación entre reproducciones sexuales y asexuales.

Background

Perennial grasses are a dominant growth form in most of the world's biodiverse, old-growth savannas. Grasslands and savannas cover between 30 and 40 percent of the Earth's surface (Blair et al. 2014). Fire, along with climate and herbivory, is considered a key driver of savanna biodiversity and persistence through time (Archibald et al. 2019). Perennial grass flammability strongly influences fire regimes that maintain tropical savanna systems (Fill et al. 2016; Pausas et al. 2017; Simpson et al. 2022; Zanzarini et al. 2022). Grass species diversity is high in tropical and subtropical savannas, and species are often locally co-dominant, varying in distribution with habitat conditions and coexisting with species-rich assemblages of forbs (Sarmiento 1983; Baruzzi et al. 2023; Siebert et al. 2024). Understanding variation in fire response and life history strategies among perennial grasses lends insight into the persistence of savannas and grasslands under a wide range of fire regimes (Solbrig et al. 1996; Enright et al. 2015; Rodrigues & Fidelis 2022; Fill & Crandall 2024).

Perennial grasses exhibit diverse ecological strategies in savannas, reflecting trade-offs among rates of growth, survival, and reproduction. Life-history theory predicts that such trade-offs shape species' responses to recurrent events such as fire by influencing how individuals allocate resources across vital rates. Fire can affect individual vital rates either directly by killing the plant or indirectly by changing the abiotic environment (Resco de Dios 2020). Fire effects on individual plants scale up to fire regime effects on population demographic rates. Collectively, the relative contributions of vital rates determine the population growth rate, which can be used as a measure of fitness (Laughlin 2023). Population growth rates of species can vary significantly between areas with contrasting fire regimes and between species with different life histories (Weekley & Menges 2012; Levick et al. 2015; Simpson et al. 2019). For example, species with strongly fire-stimulated flowering may maintain stable population growth through high adult survival (Fill & Crandall 2024), while those less dependent on fire to stimulate seed production may show variable population growth rates in years with and without fire as a result of greater sensitivity to fluctuating abiotic conditions.

Our objective was to model population growth rates of dominant perennial grasses that differ in reproductive modes in tropical and subtropical pine savannas of North and Central America. Pine savannas are old-growth ecosystems that have historically been understudied despite their presence worldwide (Myers & Rodriguez-Trejo 2009; Noss, 2013, 2015; Fill et al. 2022; Pau et al. 2023). We modeled the demography of four grass species that displayed different degrees of fire-stimulated flowering (Lamont et al. 2019), and investigated whether differences in vital rates among species influenced population growth rates in years with and without fire. We predicted that: (1) obligately fire-stimulated flowering species would show high adult survival regardless of fire and maintain stable population growth rates; (2) species with weakly or no fire-stimulated flowering would exhibit more variable survival and reproductive rates and population growth rates. By linking demographic responses to life-history theory, our study provides a framework for understanding the mechanisms that allow perennial grasses to persist under recurrent fire.

Methods

Study areas

USA:

From early 2019 to fall 2021, we monitored the annual growth, survival, and reproduction of *A. beyrichiana* (wiregrass) and *Sporobolus junceus* (pineywoods dropseed) populations in two pine savanna units (referred to as “C10A” and “C10B”) at Ordway-Swisher Biological Station, a University of Florida research property located approximately 30 km northeast of Gainesville in north-central Florida (29.41° N, 82.00° W). The climate is humid, with 20°C mean annual temperature and 1432 mm mean annual precipitation (Readle 1990). The seasonal climate alternates between hot summers with frequent thunderstorms (May–September) and cool, relatively dry winters (Fill et al. 2019). Both pine savanna units had similar habitats (i.e., xeric sandhills), vegetation structure (i.e., sparse overstory of *Pinus palustris* (longleaf pine), and limited midstory of *Quercus laevis* (turkey oak), and soil type (Candler and Apopka soils). Both units had been managed with similar fire frequency since 2003 (i.e., between 2 and 3

years), primarily during April, May, and June, and were burned at least once during this study. Before and during our study, Unit C10A was burned in April 2015 and June 2019, and Unit C10B in February 2018, March 2019, and April 2021. Thus, plants were sampled in both fire and non-fire years in both units.

Belize:

Between October 2019 and December 2021, we monitored the annual growth, survival, and reproduction of *Mesosetum filifolium* (wiregrass) and *Paspalum pectinatum* (grama-das-pedras) in Payne's Creek National Park (PCNP) in the southern coastal plain of Belize (–88.58 N, 16.3 W). The climate in Belize is subtropical to tropical, with the annual rainy season extending from May–January and the dry season lasting from February–April (McLoughlin et al. 2015). The mean annual rainfall in the south is 3800 mm (National Meteorological Service of Belize 2021). The monthly minimum temperature ranges from 16 to 28°C in winter (corresponding to the beginning of the dry season) and 24–33°C in summer (roughly corresponding to the wet season; Stuart et al. 2006). Nutrient-poor, lowland savannas in PCNP have a sparse overstory of Caribbean pine, interspersed with subcanopy trees and understory shrubs such as *Byrsonima crassifolia* (craboo), with dense groves of *Acoelorrhaphe wrightii* (palmetto palm) along riparian borders. Soils may become waterlogged during the rainy season (Blentlinger & Herrero 2020). The herbaceous understory is composed of diverse grasses, sedges, and legumes (Goodwin et al. 2013). Fire management targets a fire return interval of 3–5 years in the park (McLoughlin et al. 2015), which currently experiences a regime of nearly annual dry-season fires because of local burning practices (McLoughlin et al. 2015; Smith 2019).

We selected these locations based on several criteria. First, both sites are frequently burned pine savannas with documented fire histories, allowing us to study demographic responses under active fire management. Second, land managers at both Ordway-Swisher Biological Station and Payne's Creek National Park requested demographic studies to inform their fire management decisions and were amenable to assisting with prescribed fire applications during our study period, enabling logistical feasibility. Third, the sites represent geographically distinct pine savanna ecosystems (i.e., subtropical

Florida versus tropical Belize), allowing us to examine whether demographic patterns are consistent across different climatic conditions while maintaining similar fire-dependent ecological characteristics.

Study species

Aristida beyrichiana is a caespitose C4 perennial grass endemic to the southeastern United States from North Carolina to Mississippi (**Fig. 1A**). In the absence of fire, plants grow vegetatively via compact tiller production. Most tillers (85%) die within one year of formation (Parrott 1967). The species is highly flammable (Fill et al. 2016) and flowers primarily after fire, even after decades without fire (Shearman et al. 2019; Fill et al. 2012, 2021). Flowers and seeds are produced about six months after fires, and seeds germinate the following spring. The inflorescence is a compact panicle of spikelets with one floret each, and the lemma enclosing the seed ends in three awns (**Fig. 1C**; Hall 1989). Seed production and viability are generally found to be highest after late dry season fires (e.g., May-June; van Eerden 1997; Rodriguez 2011; Fill et al. 2012). Previous studies suggest a possible tradeoff between inflorescence production and seed viability, depending on the fire season (Baruzzi et al. 2022; Armstrong 2021). In the absence of soil disturbance, the species is long-lived, with population growth occurring via basal area expansion, fragmentation, and seed germination (Laucevicius et al. 2021).

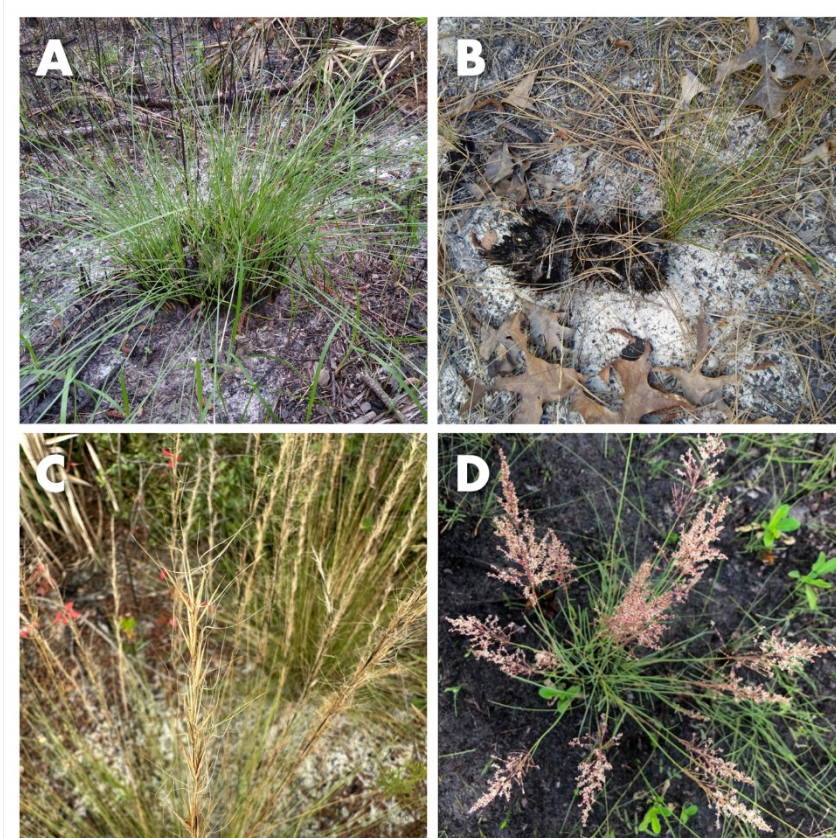


Figure 1. A) *Aristida beyrichiana* plant post-fire; B) *Sporobolus junceus* plant post-fire; C) *A. beyrichiana* inflorescences; D) *S. junceus* inflorescences

Sporobolus junceus is a C4 perennial bunchgrass that grows from Virginia to Texas (**Fig. 1B**). In its vegetative state, it is nearly identical to *A. beyrichiana*. The inflorescence is a spreading panicle with branches of scattered spikelets with no awns (**Fig. 1D**; Hall 1989). Although it can flower year-round without fire (Institute for Regional Conservation 2020), a previous study at our site found that *S. junceus* plants produced mature seeds within 7 to 9 weeks of a May fire (Canfield & Tanner 1997).

Both species exhibit little to no seed banking, with most seeds germinating within 6 months (Mulligan & Kirkman 2002; Coffey & Kirkman 2006). However, the authors and others have observed *Aristida* germinating in the second year after being sown in restoration sites. For this study, we considered

no seed bank viability. Spikelets of both species are known to be infected by smut fungi. *A. beyrichiana* flowers are infected by *Langdonia walkerae* (Alqurashi et al. 2021; Fill et al. 2024), while *Macalpinomyces sporoboli* has been found on *S. junceus* (Vánky 2003). Both species are associated with diverse understory plants (Baruzzi et al. 2022).

Mesosetum filifolium and *P. pectinatum* are dominant bunchgrass species in coastal Belizean savannas, with ranges extending to other parts of Central and South America (Fig. 2A, B; Farrugia et al. 2004). Both can flower with or without fire (S. Sylvester, personal communication). The inflorescence of *M. filifolium* is a solitary raceme, while that of *P. pectinatum* may have several racemes (Fig. 2C, D; Sylvester 2017). Almost no literature exists on *M. filifolium* life history. *Paspalum pectinatum* grows in a variety of environments throughout its distribution, including sandy savanna soils, flooded soils, riverbanks, and roadsides (Delfini et al. 2017), although most research on this species has been conducted in Brazil. In Brazil, *P. pectinatum* exhibits greater flowering post-fire (Fidelis et al. 2022; Le Stradic et al. 2015; Fontenele et al. 2020), and flowering and seed dispersal generally occur within a few months of fire or at the beginning of the rainy season; seeds do not exhibit dormancy (Fontenele et al. 2020; Fontenele & Miranda 2022). The species also show high abundance under an annual fire regime (Rodrigues & Fidelis 2022).

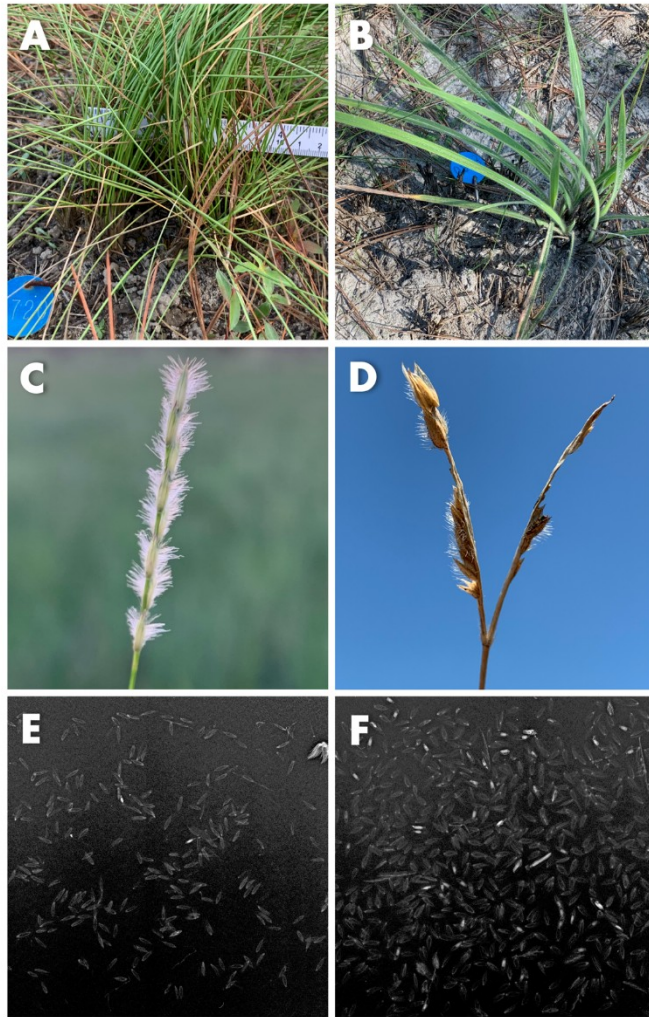


Figure 2. A) *Mesosetum filifolium* plant; B) *Paspalum pectinatum* plant; C) *M. filifolium* inflorescence; D) *P. pectinatum* inflorescence; E) X-ray image of *M. filifolium* seeds; F) X-ray image of *P. pectinatum* seeds

Data collection for vital rates

In 2019 and 2020, we selected and marked individual plants in two populations each of *A. beyrichiana* ($n_{2019}=109$, $n_{2020}=107$), *S. junceus* ($n_{2019}=108$, $n_{2020}=113$), *M. filifolium* ($n_{2019}=100$, $n_{2020}=99$),

and *P. pectinatum* ($n_{2019}=100$, $n_{2020}=100$). One *A. beyrichiana* and one *S. junceus* population were located in a single unit at the Florida site (Unit C10A), and the other population of each were located in another unit at the Florida site (Unit C10B; **Fig. 3**). Similarly in Belize, one population each of *M. filifolium* and *P. pectinatum* were located in one unit (Unit 1), and another of each in Unit 14 (**Fig. 3**). Plants were selected within each unit such that we had a representative sample from a range of sizes for the greatest chance of capturing size-dependent demographic processes. We measured plant size, survival, and fecundity for each transition (**Fig. 3**). To determine plant size, we measured the length of the longest axis and the axis perpendicular to the longest axis, calculating the basal area as the area of an ellipse. Plants were counted as dead if no green tillers were visible. If plants were not located one year but found the next, all data except survival were entered as “NA.” Fecundity measures included whether a plant was reproductive (0/1) and the number of seeds produced. We considered a plant reproductive if it had at least one flowering culm.

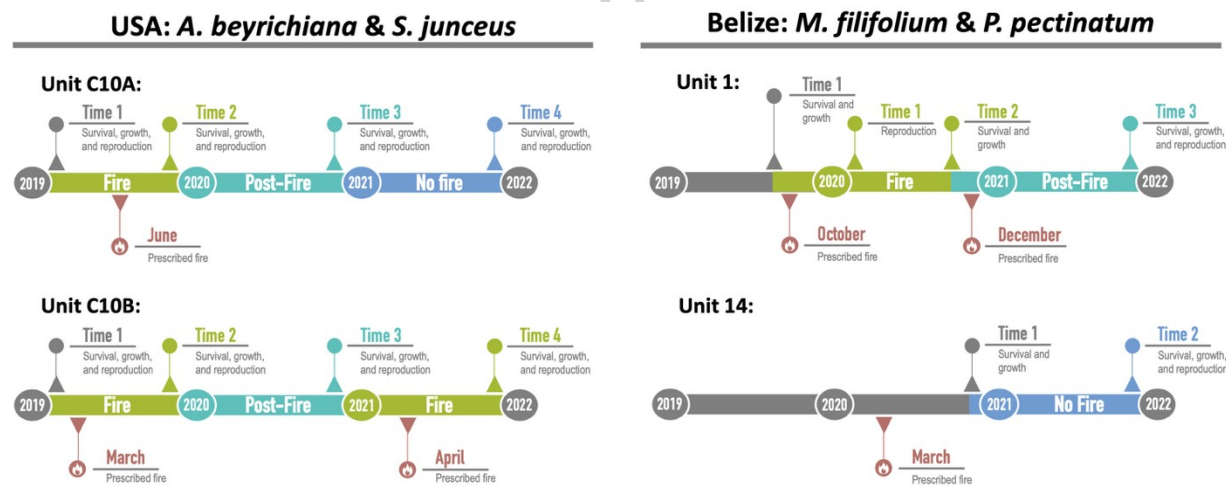


Figure 3. Timeline showing timing of fires and sampling, which correspond to different transition types (fire, post-fire, and no fire) for grasses in USA and Belize. Reproduction data were reused for several transitions due to COVID-related disruptions to sampling (See Appendix S1). The “Times” (e.g., Time 1, Time 2) indicate the beginning and end of each successive

transition, and not necessarily the sampling times, which are described in the Methods and in **Appendix S1**.

Each population was monitored for 2-3 consecutive years, with annual measurements taken in the fall after the growing season. The timing of prescribed fires varied among units and years (Figure 3), creating different transition types: (1) 'fire' transitions where the population burned during the year, (2) 'post-fire' transitions in the year immediately following fire, and (3) 'no-fire' transitions occurring ≥ 2 years after the most recent fire. This sampling design enabled us to quantify demographic responses across the full range of fire regime conditions typical of these systems.

To estimate the number of seeds, we collected and measured 20-30 or 50 flowering culms, each from a different plant during each flowering event. For *A. beyrichiana*, we counted the glume pairs of each inflorescence. We regressed the number of glume pairs on inflorescence length to obtain an equation for calculating seed number from inflorescence length for each population and transition. Specifically, we measured the inflorescence lengths of all culms on each plant and used the equation to estimate the number of seeds per plant. We followed a similar procedure for the Belize bunchgrasses. For *S. junceus*, we used a different but analogous method because the inflorescences have a different structure (Hall 1989), instead relating the number of branches on each inflorescence to the number of seeds.

To determine establishment, we searched for but found no naturally occurring seedlings in the field. To determine germination, we collected *A. beyrichiana* seeds in the fall after fires, stored them at approximately 3°C, and germinated them in the lab the following March. *Sporobolus junceus* germination for fire years was estimated from seeds collected a couple of months after fires from plants in a unit with similar fire history, which burned in June. *Paspalum pectinatum* seeds were germinated in the lab in September 2020. Photographs taken using a digital x-ray (Kubtec XPERT 80 Digital Radiography System, www.kubtec.com) indicated there were very few filled seeds of either species (**Fig. 2E, F**). Seeds

were placed on blotter paper in polystyrene boxes to which we added 5 mL of a 0.002% solution of Plant Preservative Mixture (Plant Cell Technology, Washington, USA). We placed the boxes on a lab bench at room temperature (Fill et al. 2021). We defined germination as the elongation of the radicle and epicotyl from the seed coat within three months of initiating germination trials.

Data analysis

We used integral projection models (IPMs) and bootstrapping to calculate population growth rates for each population and transition. Integral projection models use continuous, rather than discrete, relationships between size and vital rates to estimate population growth (Ellner & Rees 2006; **Appendix S1 Table S1**). We constructed separate integral projection models (IPMs) for each species-by-transition combination (e.g., *A. beyrichiana* fire transition, *A. beyrichiana* post-fire transition, etc.) rather than including fire as a covariate in a single model. This approach allows vital rate relationships to vary completely between fire regimes, which is more biologically realistic and captures potentially nonlinear effects of fire regime on demography. IPMs use continuous size distributions to model population dynamics by tracking how individuals survive, grow, reproduce, and recruit as functions of their size. For each species and transition, we modeled four vital rates: (1) survival probability as a function of plant size; (2) growth (change in size for surviving plants) as a function of current size; (3) probability of reproduction as a function of size; and (4) seed production per reproductive plant as a function of size, combined with germination rates from laboratory trials. We used bootstrapping (1000 iterations) to estimate 95% confidence intervals around population growth rates (λ). Detailed model parameterization, including functional forms, parameter estimates, model selection criteria, and diagnostics, are provided in Appendix S1 Tables S2-S9 and Appendix S2 Tables S1-S30 to ensure full reproducibility.

Results

All species exhibited stable or growing populations (**Table 1; Fig. 4**). The population growth rate was at or slightly above 1 for all species and transitions. One population of *S. junceus* had consistently

high growth rates regardless of fire occurrence (**Table 1; Fig. 4**). One *A. beyrichiana* and one *P. pectinatum* population had population growth rates above one during a post-fire year and a fire year, respectively. Even in years with no reproduction, all species maintained stable population growth rates.

Table 1. Means and confidence intervals for bootstrapped population growth rates of grass populations subject to different fire regimes (a=*A. beyrichiana*; s=*S. junceus*; m= *M. filifolium*; p= *P. pectinatum*)

Unit	Transition 1	Transition 2	Transition 3
C10A a	1.00 (1.00, 1.00)	1.08 (1.07, 1.08)	1.00 (1.00,1.00)
C10B a	1.00 (1.00, 1.00)	1.00 (1.00,1.00)	1.00 (1.00,1.00)
C10A s	1.14 (1.08, 1.19)	1.32 (1.32, 1.33)	1.26 (1.17, 1.35)
C10B s	1.00 (1.00, 1.00)	1.00 (1.00,1.00)	1.00 (1.00, 1.00)
Unit 1 m	1.00 (1.00,1.00)	1.00 (1.00,1.00)	
Unit 14 m	1.00 (1.00,1.00)		
Unit 1 p	1.01 (1.01,1.01)	1.00 (1.00,1.00)	
Unit 14 p	1.00 (1.00,1.00)		

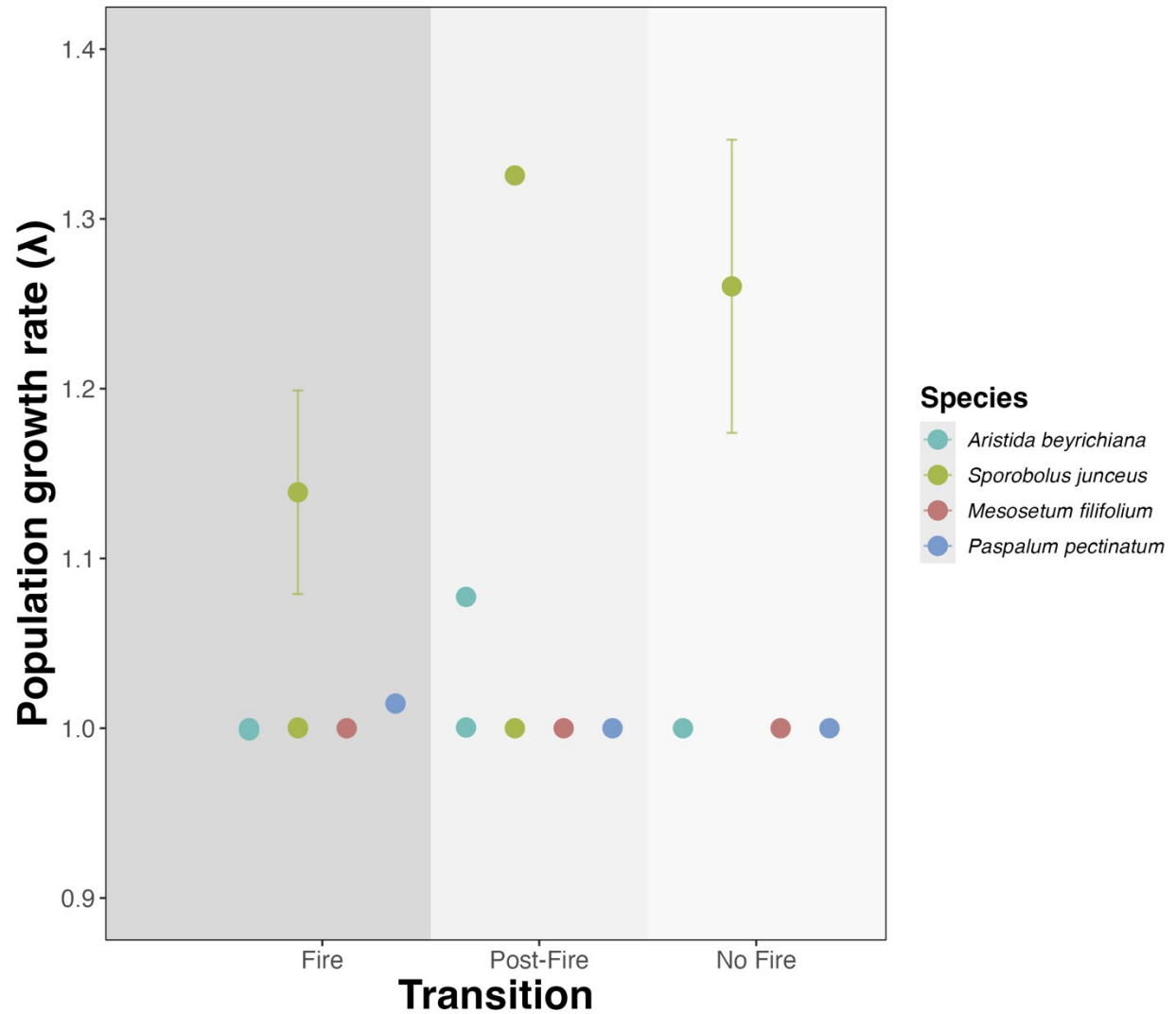


Figure 4. Population growth rates of four grass species during one-year transitions. Each point represents a population, but points overlap if growth rates are the same. *Sporobolus junceus* and *A. beyrichiana* populations were studied in the southeastern United States, and *M. filifolium* and *P. pectinatum* were studied in southern Belize. “Fire” indicates a transition in which the population was burned, “post-fire” indicates a transition following one with fire, and “no fire” indicates a transition without fire and not immediately following one with fire. Error bars are 95%

confidence intervals from bootstrapping. For most points, the confidence intervals were too small to extend beyond the plotted points.

Survival rates were consistently high across all species and transitions. Survival exceeded 95% in most transitions for all species (Appendix S1 Tables S2-S9). Growth rates varied among species and transitions, with plants generally showing slow but steady increases in basal area. The relationship between plant size and survival was generally positive, with larger plants having higher survival probability, though this relationship varied in strength among species and transitions (Appendix S2 Figs. S1-S30).

Species showed contrasting reproductive patterns across transitions. *Aristida beyrichiana* exhibited strong fire-stimulated flowering, with reproductive plants observed primarily during the fire and post-fire transitions. *Sporobolus junceus* showed more facultative flowering, producing seeds across multiple transition types but with higher seed production following fire. *Mesosetum filifolium* and *P. pectinatum* both flowered across transitions, though flowering frequency and seed production varied (Appendix S1 Tables S2-S9). Germination rates from laboratory trials were generally low for all species (typically <20%), and no recruitment from seed was observed in field populations during the study period.

The species showed different patterns balancing survival and reproduction. For *S. junceus*, variable survival (ranging from 90-98% depending on transition and population) was balanced by relatively consistent seed production across all transitions. In contrast, *A. beyrichiana* showed consistently high survival (>95%) across all transitions but concentrated reproduction in fire and post-fire years. The Belize species (*M. filifolium* and *P. pectinatum*) exhibited intermediate patterns with high survival and more consistent flowering across fire regimes. These contrasting strategies, fire-cued vs. facultative

reproduction, combined with high survival, both resulted in stable population growth rates for all fire regimes we tested.

Discussion

Despite tradeoffs in survival and seed production among species and transitions, all species showed consistently stable or slightly increasing population sizes. Variable reproduction was balanced by high survival. As predicted, the obligately fire-stimulated flowering species *A. beyrichiana* showed high survival and maintained stable population growth rates, increasing in years with seed production. In contrast, *S. junceus* maintained stable or increasing population growth rates by balancing consistency in reproductive output and relatively high seed germination rates with slightly fluctuating survival rates. Although little local life history information was available for the two Belize species, *P. pectinatum* and *M. filifolium*, their reproductive patterns appear to be more strongly influenced by fire than those of *S. junceus*. Thus, they maintained population stability via high survival in the face of variable recruitment.

Given sampling limitations, we made some assumptions about recruitment (**Appendix S1**). However, the predominance of vegetative recruitment and growth over recruitment from seed has been observed in long-lived species of many grassland and savanna systems (Eriksson, 2000; Benson & Hartnett 2006; Fontenele et al., 2020; Buisson et al. 2021; Fill & Crandall, 2024). Very low seed set or viability is characteristic of slow-growing, long-lived species in grasslands and savannas, including our study species (**Fig. 2E, F**; Snyman et al., 2013; Perez, 2014; Fidelis et al., 2021; Fill & Crandall, 2024). Moreover, high survival has been shown to be important for perennial C4 grass populations in old-growth savannas. *Themeda triandra* (red oat grass), for example, is considered a keystone species in Australian savannas and requires regular defoliation to persist (Snyman et al., 2013). Survival was the most important for the population persistence of dominant perennial C4 grasses in a humid West African savanna (Koffi et al., 2022). Although individual growth rates might be most influential for the population growth of some savanna grasses (Silva et al., 1991), the survival of many grass species is likely the most

important contributor to savanna persistence in these North and Central American savanna ecosystems (e.g., Fill & Crandall, 2024).

The perennial grasses we studied exhibit life history strategies typical of old-growth savanna species with extensive and protected bud banks. Perennial grasses may have thousands of buds protected from fire below the soil surface, translating to high individual and population survival and continuous vegetative recruitment (Hendrickson & Briske, 1997; Ott et al., 2019; Fill et al., 2025). Bud bank dynamics among species are sensitive to fire regime frequency and season, which may explain why we observed more variable vital rates in *S. junceus* (Russell et al., 2015, 2019). The bud bank also plays a key role in tradeoffs between vegetative and sexual reproduction, especially in fire-stimulated flowering species like our *A. beyrichiana* (Fill et al., 2012). Although buds also provide a resource for flowering stems, if seed recruitment is low, sexual reproduction contributes relatively little to population growth in most years. As a result, allocating buds for sexual reproduction is balanced against the benefits of survival and vegetative growth (Dalglish et al., 2012).

Long-lived grass species can persist not only in frequently-burned savannas but also in those that have been fire-suppressed for decades (Pimm, 1984; Eriksson, 2000; de Witte and Stocklin, 2010; Fill et al., 2021; Fill & Crandall, 2024), providing continuity in ecosystem function (Eriksson, 2000; Veldman et al., 2015). The presence of long-lived perennial grasses that are resistant and resilient to variation in disturbances (e.g., changes in fire regimes and climate) could buffer against short-term changes and, in turn, support long-term ecosystem persistence (Morris et al. 2008).

Availability of data and materials

Data are archived in Dryad for peer review at:

http://datadryad.org/share/XnbCaCwG_65UP4sjeHbQyl2INbUe7B-fOymPzlp4fJ4.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

JMF and RMC conceived the study and analyzed the data; JMF wrote the manuscript; all authors participated in data collection and manuscript review and editing.

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